

Hypercalciuria Associated with High Dietary Protein Intake Is Not Due to Acid Load

.

Context and Objective: Dietary intake of animal proteins is associated with an increase in urinary calcium and nephrolithiasis risk. We tested the hypothesis that the acid load imposed by dietary proteins causes this hypercalciuria. Design and Setting: In a short-term crossover metabolic study, an alkali salt was provided with a high-protein diet (HPD) to neutralize the acid load imparted by dietary proteins. Participants and Interventions: Eleven healthy volunteers were evaluated at the end of each of four phases while consuming metabolic diets with fixed calcium and sodium content. Phases 1 and 3 consisted of a control diet (CD). Phases 2 and 4 consisted of a eucaloric HPD (60 g/d animal proteins added to CD). Along with HPD in phases 2 and 4, subjects ingested 30 mEq twice daily of either potassium citrate (KCitrate, alkaline salt) or potassium chloride (KCl, control neutral salt). Results: KCitrate completely neutralized the acid load imparted by HPD (based on changes in urine pH and net acid excretion) and increased urinary citrate. Urinary calcium increased during both HPD phases compared with CD but was not significantly different between the HPD + KCl and HPD + KCitrate phases (182 ± 85 vs. 170 ± 85 mg/d; $P = 0.28$). Increased urinary saturation with respect to calcium oxalate and uric acid with HPD was abrogated by KCitrate. Conclusions: This study suggests that, at least in the short-term, mechanism(s) other than acid load account for hypercalciuria induced by HPD. The beneficial effect of KCitrate on nephrolithiasis risk with HPD is through correction of declines in urine pH and citrate.